

Lab 40 — Malware Types, Symptoms and the Seven-Step Removal Procedure

Course: CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)

Topic 07: Security — Core 2, 28% of Core 2

Exam objective: Identify malware types by behaviour and apply the CompTIA seven-step malware removal procedure in order (Core 2 objectives 2.3 and 3.3).

Goal

You study malware behaviour safely in the simulator, then learn the seven-step removal procedure in order — because the order is examinable and, more importantly, because doing step four before step three re-infects the restore points you are relying on.

What you'll produce

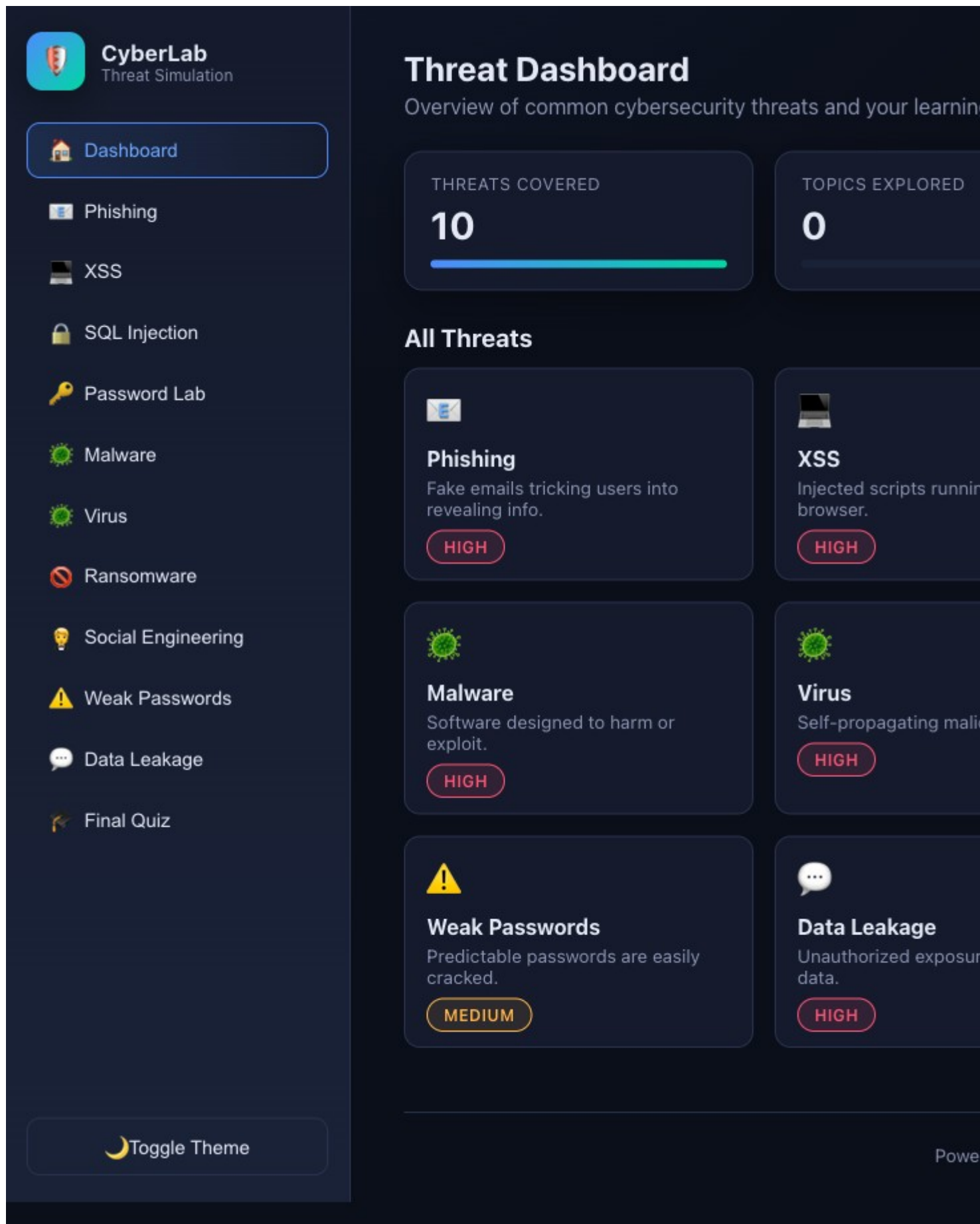
A malware type reference and a written seven-step removal procedure applied to a specific infection scenario.

Tools and equipment

Cybersecurity Simulator (<https://alfredang.github.io/cybersecuritysimulator/>) — Malware, Virus and Ransomware modules

Browser tools used in this lab

- **Cybersecurity Simulator** — <https://alfredang.github.io/cybersecuritysimulator/>



Cybersecurity Simulator — the panels and fields this lab uses.

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Open <https://alfredang.github.io/cybersecuritysimulator/> and select the Malware module, then run the visual demonstration. Nothing real executes — the animation is conceptual.
2. Run the Virus Spread Visualizer and record how propagation accelerates, then state why one infected machine on a flat network is an organisational problem rather than a user problem.
3. Open the Ransomware module and use the locked screen demonstration to see what a victim actually sees, including the countdown and the payment demand.
4. Record the ransomware response: do not pay, isolate the machine immediately, identify the variant, and restore from a backup that is offline or air-gapped.
5. Explain why an air-gapped or immutable backup is the only reliable ransomware defence: ransomware encrypts every backup it can reach over the network.
6. Build the malware reference table with rows for virus, worm, trojan, ransomware, rootkit, spyware, keylogger, cryptominer and botnet.
7. For each type record the propagation method, the primary symptom and the specific removal difficulty, and note that a rootkit may require reinstalling the operating system.
8. Record the distinguishing behaviours precisely: a virus needs a host file and human action, while a worm self-propagates across the network with no user involvement at all.
9. Write the seven-step removal procedure in order: investigate and verify symptoms; quarantine the system; disable System Restore; remediate by updating the anti-malware and scanning in safe mode; schedule scans and updates; re-enable System Restore and create a restore point; educate the end user.
10. Explain why System Restore must be disabled at step three: restore points can hold the malware, so remediating before disabling them leaves a re-infection path.
11. Explain why safe mode matters at step four: it loads a minimal driver set, so malware that hooks at startup is not running to defend itself.
12. Apply the full procedure to a scenario — a user reports pop-ups, a slow machine and a browser homepage they did not set — writing the specific action you take at each of the seven steps.

Test it — verification

Your reference table covers at least eight malware types with propagation and symptoms; the seven steps are in the correct order; you correctly explain why System Restore is disabled before remediation; and the scenario has a specific action at every step.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the <code>apt-get install</code> step at the start of the lab — the playground starts with a minimal package set.
The KillerCoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against <code>labs/tools.md</code> . All four tools run entirely client-side and need no login.
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete `worksheet.md` as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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